

**THE UNIVERSITY OF AZAD JAMMU & KASHMIR,
MUZAFFARABAD**

Department of Software Engineering

Project Report

BirdCLEF+ 2026

*Acoustic Species Identification in the Pantanal, South
America*

Deep Learning (SE-3210)

Submitted To

Engr. Ahmed Khawaja

Team Name: Team Gamma

Azan Latif 2023-SE-14

Imama Tanveer 2023-SE-28

Asghar Ali Hamza 2023-SE-42

Session: 2023-2027

Table of Contents

1. INTRODUCTION	3
2. PROBLEM STATEMENT	3
3. PROJECT OBJECTIVES	3
4. COMPETITION OVERVIEW	4
5. DATASET AND AUDIO DATA	4
6. TOOLS AND TECHNOLOGIES	5
7. METHODOLOGY	5
7.1 Audio Preprocessing	6
7.2 Audio Segmentation.....	6
7.3 Feature Extraction / Embeddings	7
7.4 Deep Learning Model	7
7.5 Prediction and Post-Processing	7
8. IMPLEMENTATION	7
9. KAGGLE TEAM.....	7
10. KAGGLE SUBMISSION AND RESULTS	8
11. RESULTS ANALYSIS	9
12. CHALLENGES AND SOLUTIONS.....	10
13. LEARNING OUTCOMES	10
14. CONCLUSION	11
15. REFERENCES	11

BirdCLEF+ 2026

1. INTRODUCTION

BirdCLEF+ 2026 is a research competition hosted on Kaggle by the Cornell Lab of Ornithology, organised under the LifeCLEF evaluation series, together with partners including the K. Lisa Yang Center for Conservation Bioacoustics, Chemnitz University of Technology, and Google DeepMind. The competition asks participants to identify wildlife species from passive acoustic recordings collected across the Pantanal wetlands of Brazil, one of the largest and most biodiverse wetland ecosystems in the world.

Bird and wildlife sound recognition is a natural fit for deep learning because it requires extracting meaningful acoustic patterns from raw audio and mapping them to species labels, a task that benefits from learned feature representations rather than manually engineered rules.

This report documents our team's participation in BirdCLEF+ 2026 as part of the Deep Learning (SE-3210) course. It covers the competition task, our understanding of the dataset, our solution approach, our Kaggle submission, and the results achieved by our team, **Team Gamma**.

2. PROBLEM STATEMENT

Given continuous audio recordings collected in the field, the objective is to predict which species are vocalising in each short segment of the recording. Because several species can call at the same time, a single audio segment may have more than one correct label, making this a multi-label classification problem rather than a single-choice one.

The recordings used for evaluation are collected under real field conditions, so a solution must remain reliable in the presence of background noise, overlapping calls, and audio characteristics that differ from the cleaner recordings typically used for training.

3. PROJECT OBJECTIVES

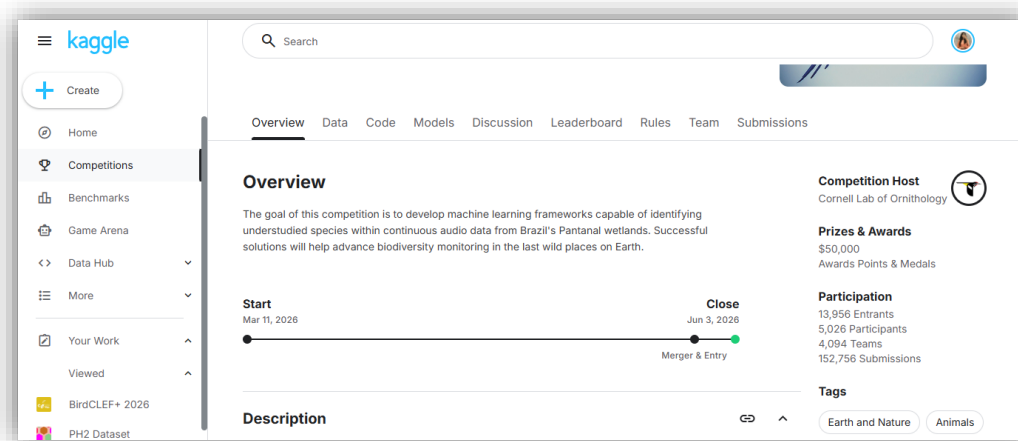
- Participate in the BirdCLEF+ 2026 Kaggle competition as a team.
- Work with real-world, field-collected bird and wildlife audio data.
- Apply deep learning and audio-processing techniques to identify species from audio.
- Generate predictions for unseen competition test recordings.
- Prepare and submit a valid entry in the format required by Kaggle.
- Evaluate performance using the competition's official leaderboard metric.
- Gain practical, team-based experience with deep learning and audio classification.

4. COMPETITION OVERVIEW

BirdCLEF+ 2026 is hosted on Kaggle as a research code competition, meaning that submissions must run entirely inside a Kaggle notebook within a fixed runtime and resource budget rather than being uploaded as pre-computed prediction files. It is organised by the Cornell Lab of Ornithology as part of the long-running BirdCLEF series under the broader LifeCLEF initiative.

Unlike earlier BirdCLEF editions that focused on birds alone, the 2026 edition extends the target species list to include other vocalising wildlife recorded in the Pantanal — such as amphibians, insects, and mammals — alongside birds, reflecting the goal of supporting broader biodiversity monitoring rather than a single taxonomic group.

Participants are given labelled training recordings and must generate per-species probability predictions for unlabelled test recordings. Kaggle scores each submission against a held-out portion of the test data and reports performance on a public and a private leaderboard, with the private leaderboard determining the final standing once the competition closes.



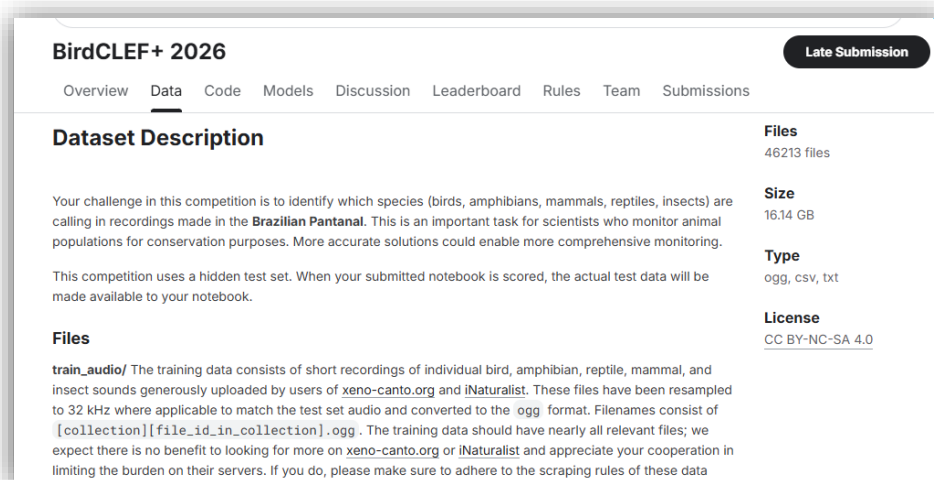
5. DATASET AND AUDIO DATA

The training data for BirdCLEF+ 2026 mainly consists of shorter, single-species focal recordings, similar in style to community-contributed recordings used in previous BirdCLEF editions, while the evaluation data consists of longer, continuous soundscape recordings captured in the field by acoustic monitoring stations across the Pantanal. This creates a domain shift between training and evaluation conditions that any solution needs to account for.

Labels in the training data are typically weak, meaning they describe which species are present somewhere in a recording rather than marking the exact moment each call occurs, so a model must learn to localise species calls in time from this coarser supervision. The competition's target species list is reported to span over 200 taxa across birds and other wildlife, several of which have very few available training examples, resulting in a long-tailed class distribution.

Component	Description
Input	Field audio recordings (soundscapes) from the Pantanal wetlands
Target	Presence of bird and other wildlife species per audio segment
Training Data	Shorter, single-species focal recordings with recording-level species labels
Test Data	Longer, continuous field soundscapes, evaluated in short segments
Output	Per-species probability predictions for each audio segment

Key challenges associated with this dataset include background environmental noise, multiple species calling simultaneously, class imbalance between common and rare species, and the mismatch between clean training audio and noisy field soundscapes used for evaluation.



6. TOOLS AND TECHNOLOGIES

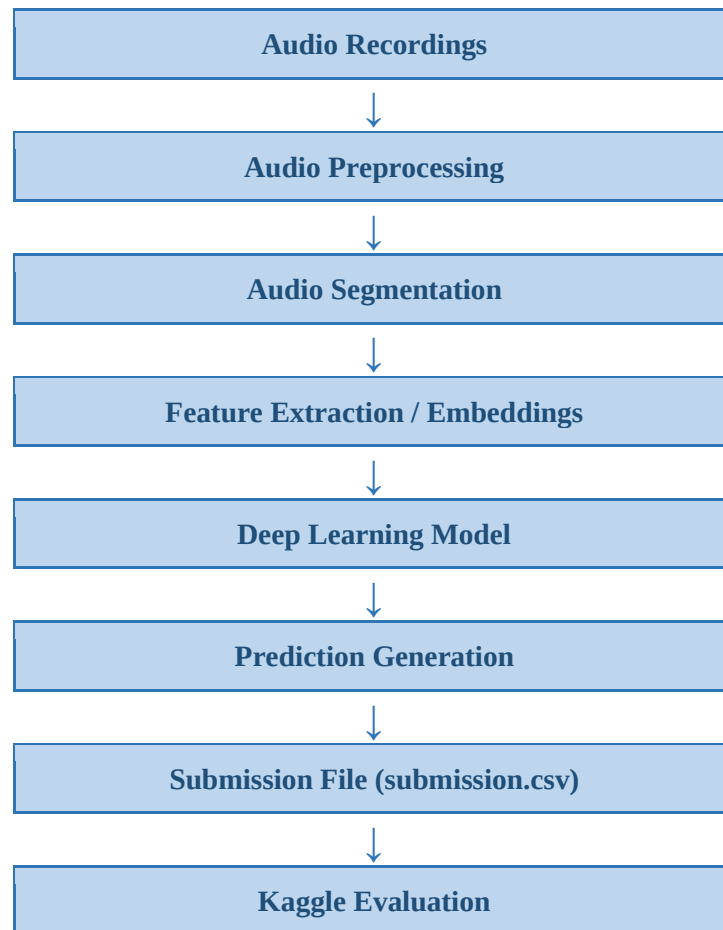
The project was carried out entirely on Kaggle, which provides the competition data, a hosted Python notebook environment, and the infrastructure used to score and rank submissions.

Tool / Technology	Purpose
Kaggle Notebooks	Hosted environment used to run the solution and generate submission files
Python	Programming language used for the implementation
Kaggle Leaderboard	Scoring and ranking of submitted predictions

7. METHODOLOGY

Solutions to acoustic species identification tasks of this kind generally follow a common processing pipeline: the input audio is preprocessed, divided into shorter segments, converted into features or embeddings, and passed through a deep learning model that outputs per-species probabilities. Figure 1 summarises this general pipeline; the exact preprocessing parameters and model configuration used in our submitted notebook are documented there.

Figure 1. Representative solution pipeline for acoustic species identification tasks



7.1 Audio Preprocessing

Before being passed to a model, raw audio is typically standardised — for example by resampling to a consistent sample rate and normalising amplitude — so that recordings from different sources and devices can be processed consistently by the same pipeline.

7.2 Audio Segmentation

Because test recordings are long, continuous soundscapes, they are divided into shorter, fixed-length windows so that a model can generate a prediction for each window rather than for an entire recording at once. This allows species presence to be estimated at a finer time resolution across a long recording.

7.3 Feature Extraction / Embeddings

A common approach in the BirdCLEF competition series is to use a pre-trained bioacoustic embedding model — such as Google's Perch models, which were developed specifically to support large-scale bird and wildlife audio classification — rather than training an audio feature extractor from scratch. Using a pre-trained model in this way is an example of transfer learning: the model has already learned general-purpose acoustic representations from a very large amount of audio, which can then be reused for a new, more specific classification task with comparatively little additional training.

7.4 Deep Learning Model

The features or embeddings produced by the preprocessing stage are passed to a classification component that maps them to the competition's species list. Depending on the approach taken, this classification stage can range from a lightweight classifier trained on top of fixed embeddings to a more complete sound event detection (SED) network trained end-to-end.

7.5 Prediction and Post-Processing

Once per-segment predictions are generated, some solutions in this competition category apply additional post-processing, such as smoothing predictions across neighbouring time segments, since a species calling in one segment is likely to still be present in the next, or combining the outputs of more than one model to obtain a more stable final prediction. The specific post-processing steps, if any, used in our final submission are recorded in our Kaggle notebook.

8. IMPLEMENTATION

The solution was implemented and executed entirely within a Kaggle Notebook using Python, in line with the code-competition format of BirdCLEF+ 2026. Predictions generated by the notebook were compiled into a submission.csv file following the column format required by the competition, and this file was submitted directly through Kaggle for scoring against the held-out test data.

Because BirdCLEF+ 2026 is a code competition, the entire prediction pipeline — from reading the raw test audio to writing the final submission file — had to run inside the Kaggle notebook environment within the platform's resource and time limits, rather than being computed offline and simply uploaded as a results file.

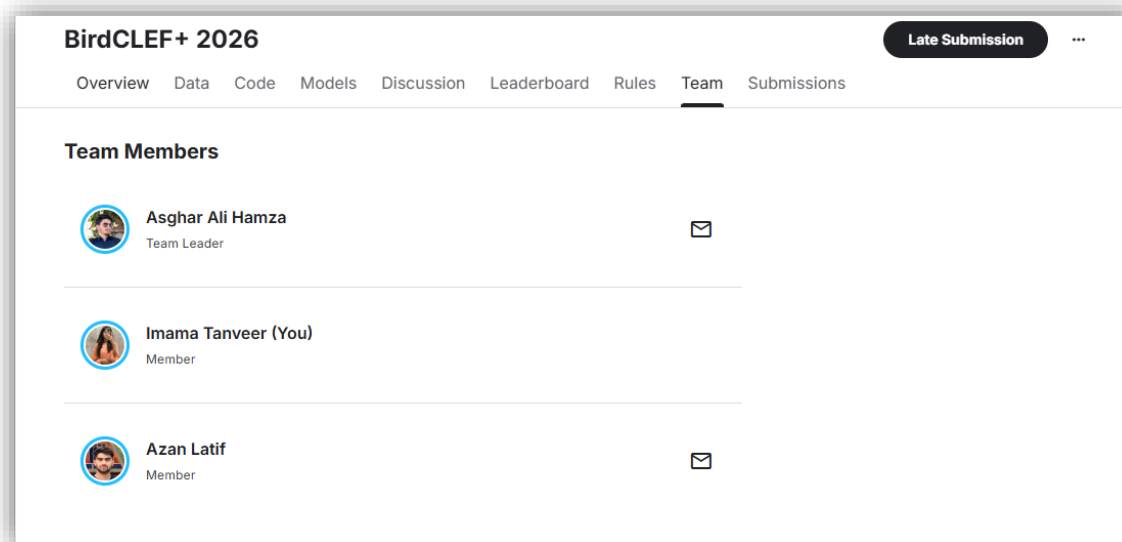
9. KAGGLE TEAM

This project was completed collaboratively by a three-member team registered on Kaggle as Team Gamma.

Member	Role
Asghar Ali Hamza	Team Leader

Imama Tanveer	Member
Azan Latif	Member

Figure 2. Kaggle Team Members – Team Gamma



As a small team, work on the competition was carried out jointly rather than being split into strictly separate, independently documented roles: notebook setup and submission, experimentation and evaluation, and reviewing results were shared responsibilities across the three members over the course of the competition.

10. KAGGLE SUBMISSION AND RESULTS

After the prediction pipeline produced a valid submission.csv file, it was uploaded to the BirdCLEF+ 2026 competition page on Kaggle. Kaggle then evaluated the submission against a held-out portion of the test data and updated the team's position on the competition leaderboard. Team Gamma submitted a total of 8 entries over the course of the competition.

KEY RESULT — TEAM GAMMA

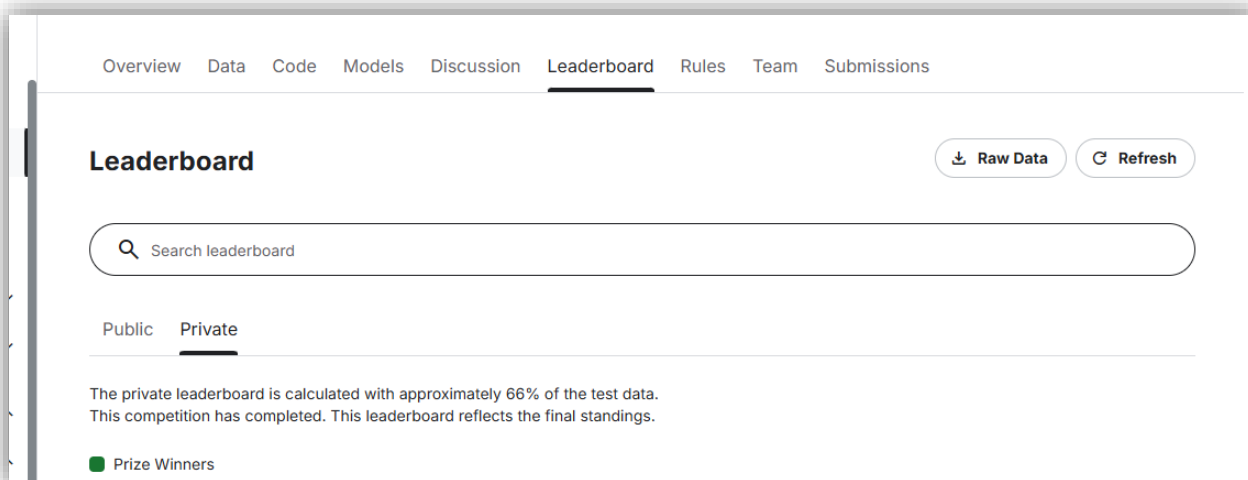
Final / Private Score: 0.94038

Final Rank: 1438

Entries: 8

Competition Status: Completed

Figure 3. Final Kaggle Private Leaderboard Result – Team Gamma



BirdCLEF+ 2026 [Late Submission](#) ...

Overview Data Code Models Discussion **Leaderboard** Rules Team Submissions

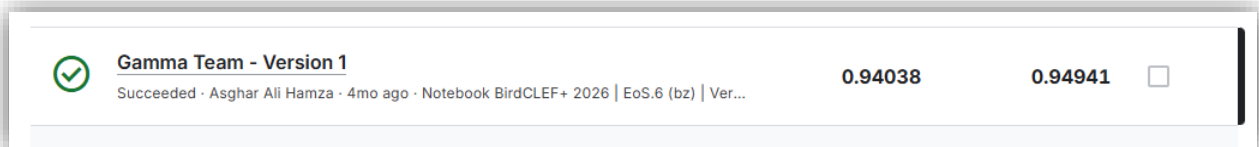
1436	▼ 156	alfaraco		0.94038	16	4mo
1437	▼ 156	JASPREET SINGH		0.94038	2	4mo
1438	▼ 154	Team Gamma		0.94038	8	3mo
1439	▼ 154	vectronyx		0.94038	10	4mo
1440	▼ 154	shibuyaboy		0.94038	12	4mo
1441	▼ 153	Wonokromo		0.94038	1	4mo

11. RESULTS ANALYSIS

The private leaderboard score of 0.94038 reflects Team Gamma's performance on the majority portion of the competition's held-out test data, and it is this score, rather than the public leaderboard score, that determines final standing once a competition is completed. A final rank of 1438 represents our team's position among all teams that submitted a scored entry to BirdCLEF+ 2026.

Across the competition, our team made 8 submissions, indicating an iterative process of testing and resubmitting entries rather than relying on a single attempt. The competition status is Completed, meaning this result reflects Team Gamma's final, official standing.

Evaluation Item	Result
Team	Team Gamma
Score	0.94038
Rank	1438
Entries	8
Status	Completed



12. CHALLENGES AND SOLUTIONS

Challenge	Approach / Solution
Audio data complexity	Standardising and segmenting recordings into consistent, model-sized windows before further processing
Computational limitations	Working within the runtime and resource limits of the Kaggle notebook environment
Model inference time	Favouring efficient inference so predictions could be generated for all test audio within the platform's time limit
Prediction generation at scale	Automating the pipeline from raw audio to a correctly formatted submission.csv file

13. LEARNING OUTCOMES

- Practical understanding of audio-based deep learning and its differences from image or text-based tasks.
- Experience with the end-to-end workflow of a real Kaggle competition, from data exploration to final submission.
- Understanding of how pre-trained audio models can be reused through transfer learning.

- Exposure to the constraints of code competitions, including notebook runtime and resource limits.
- Understanding of leaderboard-based evaluation, including the distinction between public and private scores.
- Experience collaborating as a team on a shared machine learning project.

14. CONCLUSION

This project involved our participation, as Team Gamma, in the BirdCLEF+ 2026 Kaggle competition, which addresses the problem of identifying bird and other wildlife species from field audio recordings collected in the Pantanal wetlands using deep learning techniques. Working within Kaggle's code-competition format, our team prepared and submitted a solution that produces per-species predictions for the competition's test recordings.

Our final submission, evaluated across 8 entries, achieved a private leaderboard score of 0.94038 and a final rank of 1438 in the completed competition. Beyond the numerical result, the project gave our team practical, hands-on experience with audio-based deep learning, the realities of working within a code-competition environment, and collaborative machine learning development.

15. REFERENCES

- Kaggle. BirdCLEF+ 2026 Competition Page.
<https://www.kaggle.com/competitions/birdclef-2026>
- ImageCLEF / LifeCLEF. BirdCLEF++ 2026 Task Description.
<https://www.imageclef.org/BirdCLEF2026>
- Kaggle. Competition Rules, Evaluation Metric and Leaderboard Guidelines (BirdCLEF+ 2026).
- Google. Perch — Pre-trained Bioacoustic Embedding Models, Kaggle Models.

THE END.....